Community Health & Outreach Management System (CHOMS)

Replacing paper registers with a secure, offline-first cloud platform.

MatCH (Maternal, Adolescent and Child Health Institute) | Work Integrated Learning 3 (XADAD7112w)

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The Problem & Our Solution

The Problem

Paper registers: duplicate clients, no history

Lost referrals: no outcome tracking

Delayed visibility: coordinators learn weeks later

Manual funder reports: compiled by hand

Our Solution

One cloud backend: Firebase Auth + Firestore

3 front ends: Website, PWA, Android App

Real-time dashboards: live work lists and charts

Offline-first sync: capture now, sync later

System Architecture

Presentation	HTML/CSS/JS (PWA) + Native Android (Java, MVVM, Material 3)
Business Logic	Pure JavaScript (utils.js: validation, SA ID, permissions) + Android ViewModels
Data Access	Repository Pattern (repo.js): role-scoped reads, cache, outbox, audited batch writes
Cloud	Firebase Auth, Cloud Firestore (africa-south1) and server-side Security Rules

Secure by Design

Role	Access
Admin	Full system access, user management, approve reports
Coordinator	View all data, generate and save reports
CHW (Community Health Worker)	See and edit only their assigned clients and visits
Pending	New sign-ups must be approved by an Admin; no data access until approved

Zero Unaudited Writes

Paired writes: every create, update and delete is committed in one atomic batch with an auditLogs entry

Server-enforced: Firestore rules reject the write if the audit entry is missing

Immutable: nobody, including Admins, can edit or delete an audit log entry

Accountable: who did what and when, for sensitive child health data (POPIA)

Working Without Signal

- CHWs work in areas with no internet
- App saves data to a local "outbox" (localStorage)
- Service worker caches the app shell, so the app opens offline
- Reconnect: outbox flushes automatically to Firebase (every ~60 s or on demand)
- Failed items are flagged, never silently lost
- Pending changes are shown with a "Sync Now" chip

Built for the Field and the Office

SA ID Validation: Auto-fills DOB and gender, blocks duplicates

Dashboards: Stat cards, work lists, and completed visits per CHW chart

Reports: Generate for any date range, Export CSV, Print/PDF

User Management: Approve, edit, deactivate, and delete accounts

Friendly Error Handling: Plain language messages, no technical jargon

Verified, Not Assumed

50

Jest unit tests

Business logic

21

JUnit tests

Android app

76

Security rule tests

Simulating real attacks

GitHub Actions CI pipelines automatically test code on every push

Overcoming Real-World Hurdles

User deletion without a paid Cloud Function: Built a secure admin-only client fallback that removes the profile, revokes access and logs the deletion to the audit trail

UI white space bug: Fixed CSS container constraints to allow full-width enterprise layouts

Sign-in loop: Fixed caching issues by always fetching fresh profiles from the server, so pending users cannot loop

Transforming Health Outreach

Conclusion CHOMS eliminates paper trails, prevents lost referrals, and ensures 100% data accountability.

Future Work

- SMS gateway integration for appointment reminders
- Native iOS app
- Offline-first Android UI
- Integration with provincial health systems

Our Team & Live System

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Firebase project: match-choms

Website: http://127.0.0.1:8123/index.html

Web app: http://127.0.0.1:8123/app/index.html

Firebase console: console.firebase.google.com/project/match-choms